

Fact Sheet

RUSSIAN MILITARY AND INTELLIGENCE RENDEZVOUS AND PROXIMITY OPERATIONS

June 2026

CONTACT: VICTORIA SAMSON
VSAMSON@SWFOUND.ORG



SECURE
WORLD
FOUNDATION

SUMMARY

Since 2013, Russia has conducted robotic rendezvous and proximity operations (RPO) in both low Earth orbit (LEO) and geostationary Earth orbit (GEO), most of which are related to military or intelligence operations. Most of these missions have been publicly acknowledged by the Russian government but few have detailed public information available. None of the programs listed here are thought to involve co-orbital ASAT testing or deployment, but there are still suspicions surrounding several of these events and concerns that some of these programs are intended to develop co-orbital counterspace capabilities.

RENDEZVOUS AND PROXIMITY OPERATIONS

Proximity operations are a series of orbital maneuvers executed to place and maintain a spacecraft in the vicinity of another space object on a relative planned path for a specific time duration to accomplish mission objectives. Rendezvous is a process whereby two space objects, artificial or natural bodies, are intentionally brought close together through a series of orbital maneuvers at a planned time and place. Taken together, RPO technologies enable a wide range of capabilities to support civil and commercial space activities such as on-orbit inspections, repair, refueling, assembly, and life extension. RPO capabilities can also be used for military and intelligence space activities such as intelligence, surveillance, and offensive weapons such as co-orbital anti-satellites. Over the past decade, Russia has conducted multiple missions practicing close operations and gathering communication intelligence. Most of these missions are suspected to be part of the Nivelir program aiming to strengthen intelligence operations and space situational awareness capabilities.

RUSSIAN RPO FOR MILITARY AND INTELLIGENCE MISSIONS IN LEO

The first known event was on December 25, 2013, when a Russian Rockot launch vehicle from Plesetsk placed three small satellites into LEO in what appeared to be another routine launch to replenish the Rodnik constellation but also included an unannounced fourth payload, Cosmos 2491.¹ From launch through the end of 2019, Cosmos 2491 did not make any significant changes to its orbit and remained in a relatively high LEO altitude of 1500 km. On December 23, 2019, Cosmos 2491 did make a small maneuver of approximately 1.5 m/s, which was accompanied by the release of 30 pieces of orbital debris.² Given the relatively low energy of the event, it is likely that the propulsion system of Cosmos 2491 failed immediately after launch and the orbital change and fragmentation event was caused by the explosive release of the residual fuel.

Once again, on May 23, 2014, the Russian government publicly declared that another Rockot launch carried three military satellites; yet, two days later, hobbyist satellite observers indicated that a fourth payload (Cosmos 2499) was on the launch. By mid-June, hobbyists reported that Cosmos 2499 had begun a series of maneuvers to match orbits with the Briz-KM upper stage that placed it in orbit.³ The

process took several months, and it was not until the end of November that Cosmos 2499 passed within a kilometer of the Briz-KM.⁴ Amateur radio operators also reported that Cosmos 2499 appeared to be using the same radio frequencies as Cosmos 2491, suggesting they used the same Yubileiny-2 microsatellite bus.⁵ On March 26, 2016, Cosmos 2499 made another orbit adjustment that slowly brought it closer to the Briz-KM by about tens of kilometers per day in order to perform proximity operations.

On March 31, 2015, a third Rockot launch took place from Plesetsk with what was publicly declared as carrying three GonetsM satellites and a classified military payload (Cosmos 2504).⁶ Cosmos 2504 began a small series of maneuvers in early April to bring it close to the Briz-KM upper stage that placed it in orbit. At some point during that pass, the Briz-KM's orbit was disturbed by an unknown perturbation, which could have been the result of a minor collision between the two space objects. If it was, the impact was very slight and did not result in additional debris being generated. On July 3, 2015, Cosmos 2504 made another significant maneuver, lowering both its apogee and perigee significantly by around 50 km each, further separating itself from the Briz-KM. On March 27, 2017, after more than a year of dormancy, Cosmos 2504 made a series of maneuvers that lowered its orbit, and on April 20, it passed within two kilometers of a piece of Chinese space debris from their 2007 ASAT test.⁷ This suggests that Cosmos 2504 has a satellite inspection or observation mission. Cosmos 2504 maneuvered again on December 10, 2019, to lower its perigee by 40 km, although the reason is not yet known.⁸

On November 25, 2019, Russia conducted another launch of a Soyuz-2-1v from Plesetsk with Cosmos 2542 on-board to conduct space surveillance as well as Earth remote sensing.⁹ On December 6, Cosmos 2542 released a small subsatellite (Cosmos 2543) that was publicly announced by Russia.¹⁰ Cosmos 2543 remained within 2 km of Cosmos 2542 for three days before it conducted a series of maneuvers to raise its apogee to 590 km by December 16.¹¹ Subsequent analysis by amateur observers strongly suggests that the purpose of these maneuvers was to place Cosmos 2543 in an orbit where it could observe a classified US intelligence satellite, USA 245, which was launched in 2013 and is believed to be an electro-optical imagery satellite operated by the National Reconnaissance Office. The orbits of Cosmos 2543 and USA 245 are synchronized such that Cosmos 2543 came within 20 km of USA 245 several times in January 2020 and since then periodically comes within 150 to 300 km of USA 245 while the latter is illuminated by the Sun; it can observe both sides of USA 245 continuously for up to a week at a time.¹²

Additionally, in June through October 2020, Cosmos 2543 rendezvoused with Cosmos 2535 which was launched in 2019 alongside four other payloads (Cosmos 2536, Cosmos 2537, and Cosmos 2538) and released a small object at high relative velocity, causing US Space Command to call it a "non-destructive test of a space-based anti-satellite weapon."¹³ Cosmos 2535 and Cosmos 2543 remained in close proximity through August 2020, and by August 13, they were joined by Cosmos 2536. Cosmos 2536 and Cosmos 2535 remained in close proximity to one another again in September 2020 until October, when Cosmos 2536 maneuvered approximately 20 kilometers away, before approaching once more nine days later. This back and forth movement continued into 2021, with reports indicating Cosmos 2536 docked with Cosmos 2535 in March 2021. There is some speculation as to the nature of Cosmos 2535 and Cosmos 2536 due to unexplained particulate clouds and orbital debris generated by their RPO activities. These activities generated by Cosmos 2535 and Cosmos 2536 indicate a possible co-orbital ASAT.¹⁴

In August 2022, Russia launched Cosmos 2558 in LEO. The behavior of the satellite as well as details of its launch seem to be oriented towards shadowing USA 326, a classified NRO imagery satellite.¹⁵ Cosmos 2558 launched into a similar orbital plane as USA 326 and periodically came within 60-70 km, although not close and synchronized enough to be a formal RPO. On August 18, 2022, USSPACECOM released a statement condemning Russia for this behavior, calling the activities of Cosmos 2558 "dangerous and

irresponsible behavior.”¹⁶ In June 2025, Cosmos 2558 released an object, later called Object C.¹⁷ Cosmos 2558 is thought to be part of the Nivelir inspector satellite program, as it has been in the orbital plane of USA 326 since its launch. Object C was observed to be maneuvering in respect to Cosmos 2558 through the summer of 2025; additionally, Object C lowered its orbit in July 2025 so it would frequently get within 100 km of USA 326.¹⁸

Cosmos 2565 was launched on December 1, 2022, and put into LEO;¹⁹ two days later, it released a small satellite, Cosmos 2566. In late December, a third object, labeled Object D, was released from Cosmos 2566. On October 27, 2023, Cosmos 2570 was launched into a similar orbital profile. It also released another satellite, Cosmos 2571, after circularizing, which once again released a third satellite, cataloged as Object D, in late November. Commercial satellite tracking firm LeoLabs reported that Object D had maneuvered and performed at least one close pass of Cosmos 2571, potentially as part of an imaging attempt.²⁰

Three satellites were launched in February 2025 on a Soyuz-2.1V launch vehicle from Plesetsk: Cosmos 2581, Cosmos 2582, and Cosmos 2583.²¹ Their given mission was just that they were for the “Ministry of Defense.”²² Originally, Cosmos 2581 was put in an orbit of 595 x 578 km; Cosmos 2582, 597 x 576 km orbit; and Cosmos 2583, 597 x 580 km.²³ Cosmos 2581 and Cosmos 2582 started conducting RPOs, at times getting less than 1 km from each other, with Cosmos 2582 moving and Cosmos 2581 being the recipient of the RPOs.²⁴ In March 2025, Cosmos 2583 released “Object F” and started conducting RPOs with it.²⁵ Cosmos 2583 maneuvered away from Object F in May 2025 to get close to Cosmos 2581 (at times, getting within several hundred meters in early June 2025); they were joined by Cosmos 2582 in late June 2025.²⁶ These RPOs continued throughout the year; US commercial SSA company COMPSOC released data that showed Cosmos 2581 and Cosmos 2583 conducting RPOs in October 2025, getting within 80 m of each other at least once.²⁷

RUSSIAN RPO FOR MILITARY AND INTELLIGENCE MISSIONS IN GEO

Russian robotic RPO activities have also occurred in GEO. On September 28, 2014, a satellite built for the Russian Ministry of Defense and Federal Security Service (FSB) was launched. The name of the satellite is not precisely known, yet often referred to as Luch according to Russian filings with the United Nations or “Olymp” or “Olymp-K” in manufacturer documents.²⁸

Over the next several months, Luch conducted a series of maneuvers that brought it close to other operational satellites around the GEO belt. During its time on orbit, Luch parked near more than two dozen commercial communications satellites for periods ranging from a few weeks to nine months,²⁹ and typically close enough to be within the typical ground terminal uplink window.³⁰ Luch also parked next to the French-Italian Athena-Fidus military communications satellite, prompting accusations of “espionage” from the French Minister of Defense.³¹ In February 2019, Luch approached the Chinese TJS-3 satellite within 30 kilometers several weeks after TJS-3 was launched. This was the first known event between a maneuverable Russian satellite and a maneuverable Chinese satellite.³² Luch’s closest known approach was within 1.8 km of Intelsat 36.³³ It was put in a graveyard orbit in October 2025;³⁴ prior to that, the last time it had been seen maneuvering was in August 2022. In January 2026, a Swiss SSA company posted videos of Luch undergoing a fragmentation event and tumbling.³⁵ It is unclear what the cause was; possibilities include an impact with a piece of debris or non-passivated fuel sources exploding.³⁶ It was equipped with KM-60 Hall-effect ion thrusters that were known to be designed to be ignited 8,250 times and operate over 4000 hours.³⁷

The orbital history of Luch before it was placed into a graveyard orbit is documented in Figure 1.

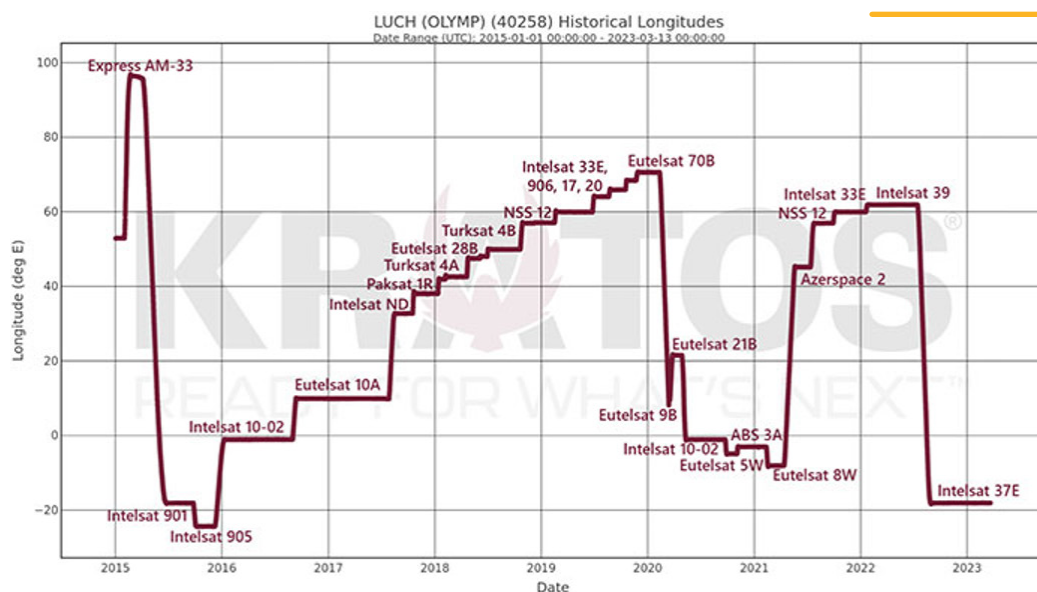


Figure 1: A compilation of Luch's orbital history and satellites visited before it was placed in a graveyard orbit in October 2025.

Credit: Kratos

In October 2022, Russia launched Cosmos 2561 and Cosmos 2562, which may be part of the Numizmat program (which was started in 2014 and appears to involve the development of a space-based ultra-wideband (UWB) radar payload).³⁸ In November 2022, a formerly inactive Russian Earth observation satellite, Resurs-P3, performed a large maneuver to drop its orbit by 20 km, and consequently, Cosmos 2562 maneuvered to conduct an RPO of Resurs-P3, possibly to collect imagery and conduct an inspection to determine the cause of its failure.³⁹

In March 2023, a Russian Proton-M rocket from Baikonur Cosmodrome placed another satellite in GEO that has since behaved very similar to Luch.⁴⁰ This new satellite, cataloged as Luch Olymp 2, and designated Luch 5X in registration filings with the United Nations, appears to be conducting RPO activities similar to that of the original Luch by visiting Russian, United States, and European satellites in the GEO belt.⁴¹ Its orbital history is documented in Figure 2 and it is still active in GEO.

Open source research has revealed more details about the originals of the Luch Olymp program. The ground-based RF tracking firm Kratos conducted an analysis of the activities of both Luch Olymp satellites and concluded they are likely being used for a signals intelligence mission, which is done by parking close enough to a target satellite to collect RF signals uplinked to it.⁴² This conclusion is corroborated by sources from within the Russian space agency who have stated that the satellite is called Olimp and had been ordered by the FSB for a signals intelligence mission, which it accomplishes by parking close enough to a target satellite to intercept uplinked radiofrequency signals.⁴³

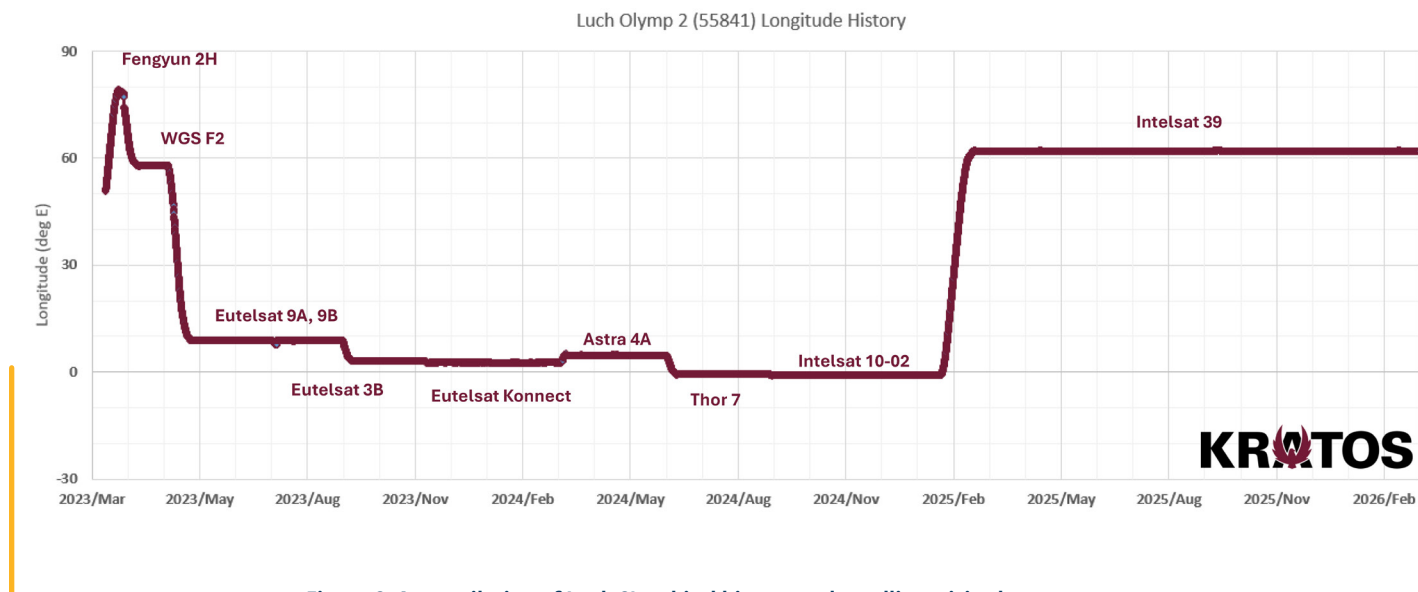


Figure 2: A compilation of Luch 2's orbital history and satellites visited.

Credit: Kratos

Cosmos 2589 was launched in June 2025 from Plesetsk Cosmodrome via an Angara A5 rocket, the first flight of the rocket with real payloads; this was after over 30 years of development, and put into a HEO orbit around the equator with a perigee of about 20,000 km and an apogee of about 50,000 km.⁴⁴ A few days later, an object was seen separating from Cosmos 2589;⁴⁵ it was later called Object C and then eventually catalogued as Cosmos 2590.⁴⁶ Cosmos 2589 and Cosmos 2590 conducted RPOs through July 2025, at one point getting less than 1 km from each other.⁴⁷ COMSPOC released a video of the two conducting RPOs in July 2025.⁴⁸ In early August 2025, Cosmos 2590 maneuvered and the two satellites pulled away from each other over the first part of the month and stopped conducting RPOs; by the end of the month, Cosmos 2590 had maneuvered enough so that the two could conduct RPOs.⁴⁹ Cosmos 2589 and Cosmos 2590 conducted RPOs in the first half of November staying within two kilometers of each other for nearly two weeks; Cosmos 2590 was the chaser and Cosmos 2589 was the target, as had been their norm.⁵⁰ In mid-November 2025, Cosmos 2589 had started conducting maneuvers to circularize its orbit, which would eventually allow it to join the GEO belt; Cosmos 2590 stopped maneuvering in early December 2025.⁵¹

DUAL-USE CAPABILITIES

The activities of these satellites are consistent with the demonstration of RPO technologies for the purpose of satellite servicing, space situational awareness, and inspection. They could also be used for aggressive purposes such as co-orbital ASATs. The lack of transparency about these programs and occasional generation of orbital debris has fueled suspicion that they are covert weapons programs. However, the publicly-available evidence to date suggests they are aimed at either broad technology demonstration or military surveillance/reconnaissance missions.

Summary of Known or Suspected Russian Military and Intelligence RPO Activities in Space

Date	Target Satellite	Chaser Satellite	Notes
Jun. 2014 - Mar. 2016	Briz-KM R/B	Cosmos 2499	Series of maneuvers to approach and retreat from the Briz-KM upper stage.
Oct. 2014 - Oct. 2025	Multiple	Luch (Olymp)	Parked near more than two dozen Russian and non-Russian satellites before being put in a graveyard orbit in October 2025.
Apr. 2015 - Apr. 2017	Briz-KM R/B	Cosmos 2504	Maneuvers to approach the Briz-KM upper stage; may have made a slight impact before separating again
Mar. - Apr. 2017	FY-1C Debris	Cosmos 2504	Close approach with debris from China's 2007 ASAT test.
Aug. - Oct. 2017	Cosmos 2519, Cosmos 2523	Cosmos 2521	Cosmos 2521 separated from Cosmos 2519 and performed a series of small maneuvers to do inspections before redocking with Cosmos 2519. Cosmos 2523 separated from Cosmos 2521 but did not maneuver on its own.
Mar. - Apr. 2018	Cosmos 2519	Cosmos 2521	Cosmos 2521 conducted close approaches of Cosmos 2519.
Aug. 2019 - Mar. 2021	Cosmos 2536	Cosmos 2535	Cosmos 2535 and Cosmos 2536 conducted at least 25 individual RPO operations to within 2 km and as far apart as 380 km.
Dec. 2019 - Mar. 2020	USA 245	Cosmos 2543	Cosmos 2543 was deployed from Cosmos 2542 and got within 30 km of USA 245, and then had repeated close approaches with USA 245 of up to 150 km.
Jun. - Oct. 2020	Cosmos 2535	Cosmos 2543, Cosmos 2536	Cosmos 2543 rendezvoused with Cosmos 2535 and released a small object at high relative velocity. In Sept., Cosmos 2536 joined in the RPO with the other two and may have docked with Cosmos 2535.
Feb. - Sep. 2022	USA 326	Cosmos 2558	Cosmos 2558 periodically came within 60-70 km of USA 326.
Nov. 20, 2022	Resurs-P3	Cosmos 2562	A formerly inactive Russian Earth observation satellite, Resurs-P3, performed a large maneuver to drop its orbit by 20 km, and on November 22, Cosmos 2562 maneuvered to conduct an RPO of Resurs-P3.
November 2023	Cosmos 2571	Object D	Cosmos 2570 released another satellite, Cosmos 2571. After circularizing, Cosmos 2571 once again released a third satellite, catalogued as Object D in late November. Object D maneuvered and performed at least one close pass of Cosmos 2571.
Mar. 2023 - Current	Multiple	Luch (Olymp) 2	Luch 2 has parked next to multiple Russian, American, and European satellites.

Date	Target Satellite	Chaser Satellite	Notes
February 2025 - Current?	Cosmos 2581, Cosmos 2582, Cosmos 2583, Object F	Cosmos 2581, Cosmos 2582, Cosmos 2583	Cosmos 2581 and Cosmos 2582 started conducting RPOs after launch, at times getting less than 1 km from each other, with Cosmos 2582 moving and Cosmos 2581 being the recipient of the RPOs. In March 2025, Cosmos 2583 released "Object F" and started conducting RPOs with it. Cosmos 2583 maneuvered away from Object F in May 2025 to get close to Cosmos 2581 (at times, getting within several hundred meters in early June 2025); they were joined by Cosmos 2582 in late June 2025. These RPOs continued throughout the year. Cosmos 2581 and Cosmos 2583 conducted RPOs in October 2025, getting within 80 m of each other at least once.
Summer 2025	Object C, USA 326	Cosmos 2558	In June 2025, Cosmos 2558 released an object, later called Object C. Object C was observed to be maneuvering in respect to Cosmos 2558 through the summer of 2025 (Cosmos 2558 had not maneuvered since April 2025). By July 2025, Object C was conducting close approaches to USA 326, frequently getting within 100 km of it.
Jun. 2025 – Nov. 2025	Cosmos 2589	Cosmos 2590	Cosmos 2589 released an object that later was catalogued as Cosmos 2590. Cosmos 2590 conducted RPOs around Cosmos 2589.

ENDNOTES

1. Brian Weeden, "Dancing in the Dark Redux: Recent Russian Rendezvous and Proximity Operations in Space," *The Space Review*, October 5, 2015, <http://www.thespacereview.com/article/2839/1>.
2. Jonathan McDowell, "Unlike the two later sats, Kosmos-2491 did not change its orbit," *Twitter*, January 12, 2020, <https://twitter.com/planet4589/status/1216265783644389376?s=20>. Total amount of orbital debris derived from the public US military satellite catalog at <https://space-track.org>.
3. Thread at the Novosti Kosmonavtiki forums, dated May 16, 2014, http://novosti-kosmonavtiki.ru/forum/forum12/topic14232/?PAGEN_1=5.
4. Posting on the Novosti Kosmonavtiki forums, dated November 28, 2014, <http://novosti-kosmonavtiki.ru/forum/messages/forum12/topic14778/message1315049/#message1315049>.
5. Пашков, Дмитрий, "Cosmos-2491/RS-46 (R4UAB)," *YouTube.com*, December 2, 2014, <https://www.youtube.com/watch?v=jHKOsdhMVDk#t=14>. The Russian government publicly disclosed the existence of the amateur radio payloads, which were activated at the end of the main mission.
6. "Rockot," *Astronautix*, accessed July 7, 2020, <http://www.astronautix.com/r/rokot.html>.
7. Anatoly Zak, "Russia Goes Ahead with Anti-Satellite System," *Russian Space Web*, updated December 15, 2017, <http://www.russianspaceweb.com/Cosmos-2504.html>.
8. Gwiz posting to the NASASpaceflight.com forums, December 11, 2019, <https://forum.nasaspaceflight.com/index.php?topic=32816.msg2024319#msg2024319>.
9. Успешный пуск ракеты-носителя «Союз-2.1в," *TASS*, November 26, 2019, <https://www.roscosmos.ru/27793/>.
10. "Минобороны провело в космосе эксперимент по отделению малого спутника от другого аппарата," *TASS*, December 6, 2019, <https://tass.ru/armiya-iopk/7285111>.
11. Jonathan McDowell, "Space Activities in 2019," January 12, 2020, pp. 29, <https://planet4589.org/space/papers/space19.pdf>.
12. Initial observations and analysis were developed by multiple observers on the See-Sat mailing list as documented here <http://www.satobs.org/seesat/Dec-2019/0108.html>. Additional analysis provided by Michael Thompson in a tweet thread posted January 30, 2020, https://twitter.com/M_R_Thomp/status/1222990126650994698. Further analysis and by Jonathan McDowell in a tweet thread posted February 1, 2020, <https://twitter.com/planet4589/status/1223420130576818176?s=20>.
13. Anatoly Zak, "Soyuz-2-1v launches four classified payloads," *Russian Space Web*, March 13, 2021, <https://www.russianspaceweb.com/Cosmos-2535-2536-2537-2538.html>; "Russia conducts space-based anti-satellite weapons test," US Space Command Public Affairs Office, July 23, 2020, <https://www.spacecom.mil/Newsroom/News/Article-Display/Article/2285098/russia-conducts-space-based-anti-satellite-weapons-test/>.
14. See Secure World Foundation's "Russian Co-Orbital Anti-Satellite Testing" fact sheet for more information: <https://www.swfound.org/publications-and-reports/2026-global-counterspace-capabilities-report>.
15. Scott Tilley (@coastal8049), "Nico Janssen noted that the NRO Keyhole USA 326 pursued by COSMOS 2558 had changed its orbit. We know that historically that KH satellites will typically use their S-band transmitters around orbital changes so I looked for it today and it appears I was rewarded!" *Twitter*, April 9, 2023, <https://x.com/coastal8049/status/1645197516395397120?s=20>.
16. Brett Tingley, "Pentagon space chief condemns 'irresponsible' launch of Russian inspector satellite," *Space.com*, August 18, 2022, <https://www.space.com/russia-inspector-satellite-kosmos-2558-irresponsible-behavior>.
17. Marco Langbroek, "Kosmos 2558 released an object on orbit on June 26 [UPDATED]," *SatTrackCam Leiden (b)log*, June 29, 2025, <https://sattrackcam.blogspot.com/2025/06/kosmos-2558-released-object-on-orbit-on.html>.
18. Langbroek, "Kosmos 2558 released an object," *ibid*.
19. Anatoly Zak, "Soyuz launches military payload," *Russian Space Web*, November 25, 2023, <https://www.russianspaceweb.com/lotos-s1-807.html>.
20. Theresa Hitchens, "LeoLabs: Russia, China time suspicious satellite activity for some US holidays," *Breaking Defense*, December 12, 2023, <https://breakingdefense.com/2023/12/leolabs-russia-china-time-suspicious-space-activity-for-some-us-holidays/>.

21. Andrew Jones, "Secretive Russian military satellites release mystery object into orbit," *Space.com*, April 3, 2025, <https://www.space.com/space-exploration/satellites/secretive-russian-military-satellites-release-mystery-object-into-orbit>.
22. Greg Gillinger, "Checking In With Cosmos 2581/2582/2583," *The Integrity Flash Issue 133*, November 10, 2025, <https://isruniversity.com/2025/11/10/issue-133/#section1>.
23. Marco Langbroek, "RPO Galore: the complex proximity operations of Kosmos 2581, Kosmos 2582, Kosmos 2583 and 'Object F'," *SatTrackCam Leiden (b)log*, June 29, 2025, <https://sattrackcam.blogspot.com/2025/06/rpo-galore-complex-proximity-operations.html>.
24. Langbroek, "RPO Galore," *ibid*. Great breakdowns of the moves, complete with diagrams, are available here: Greg Gillinger, "Another Look At Cosmos 2581, 2582 & 2583," *The Integrity Flash Issue 118*, April 8, 2025, <https://isruniversity.com/2025/04/07/issue-118/#section1>; and Greg Gillinger, "Dance-Off in LEO: Cosmos 2581 and 2582," *The Integrity Flash Issue 116*, March 9, 2025, <https://isruniversity.com/wp-content/uploads/2025/03/116-9-Mar-2025-Integrity-Flash.pdf>.
25. Langbroek, "RPO Galore," *ibid*.
26. Langbroek, "RPO Galore," *ibid*.
27. Greg Gillinger, "Checking In With Cosmos 2581/2582/2583," *ibid*.
28. Anatoly Zak, "Proton Successfully Returns to Flight Delivering a Secret Olymp Satellite," *Russian Space Web*, October 19, 2015, <http://www.russianspaceweb.com/olymp.html>.
29. Marco Langbroek, "LUCH (Olymp-K), an eavesdropping SIGINT snooping around commercial comsats," *SatTrackCam Leiden (b)log*, April 6, 2021, <https://sattrackcam.blogspot.com/2021/04/luch-olymp-k.html/>
30. Bob Hall, "Luch Space Activities," AGI web series, Ep. 14, June 26, 2019, <https://youtu.be/D67dg9P3eDY>.
31. John Leicester and Sylvie Corbet, "Espionage: French defense head charges Russia of dangerous games in space," *DefenseNews*, September 7, 2018, <https://www.defensenews.com/space/2018/09/07/espionage-french-defense-head-charges-russia-of-dangerousgames-in-space/>.
32. David Chen & Peter Singer, "'Fast movers': Chinese satellites zoom around for inspections—or interference," *Defense One*, May 20, 2024, <https://www.defenseone.com/threats/2024/05/fast-movers-meet-chinese-satellites-zoom-around-inspectionsor-interference/396714/>.
33. Hall, "Luch Space Activities," *ibid*.
34. Slingshot Aerospace, "Slingshot Orbital Alert: Russia's Luch (Olymp) has parked in Graveyard Orbit," *LinkedIn*, October 29, 2025, https://www.linkedin.com/posts/slingshot-aerospace_slingshot-orbital-alert-russias-luch-activity-7389344239664459776-C73Y/.
35. s2a systems, "LUCH (OLYMP) has started to tumble in its graveyard orbit tonight," *Twitter*, January 30, 2026, https://x.com/s2a_systems/status/2017162450001420602.
36. Andrew Jones, "Russian 'inspector' satellite appears to break apart in orbit, raising debris concerns," *Space.com*, January 30, 2026, <https://www.space.com/space-exploration/launches-spacecraft/russian-inspector-satellite-appears-to-break-apart-in-orbit-raising-debris-concerns>.
37. Bart Hendrickx, "Re: Proton-M/Briz-M launch with Luch (aka Olymp) - 2023UTC September 27, 2014" *NASASpaceflightForums.com*, April 10, 2025, <https://forum.nasaspaceflight.com/index.php?PHPSESSID=mth-3l3pmgs5u49fmgvflqkbj7c&topic=34315.msg2678514#msg2678514>.
38. Bart Hendrickx, "Re: Soyuz-2.1v/Volga - Kosmos 2561/2562 - Plesetsk 43/4 - 21 Oct 2022 19:20 UTC," *NASASpaceflightForums.com*, January 9, 2023, [<https://forum.nasaspaceflight.com/index.php?topic=57374.msg2447737#msg2447737>](<https://forum.nasaspaceflight.com/index.php?topic=573>
39. Sandra Erwin "LeoLabs data shows on-orbit maneuvers by Russian satellites," *SpaceNews*, November 6, 2023, <https://spacenews.com/leolabs-data-shows-on-orbit-maneuvers-by-russian-satellites/>.
40. Justin Mooney, "Russian Proton-M launches Olymp-K-2 military satellite," *NASASpaceflight.com*, March 12, 2023, <https://www.nasaspaceflight.com/2023/03/proton-olymp-k-2/>.
41. Brian Weeden, "Our Satellite Dashboard shows this new Luch hung out near the Space Force WGS F2 near 60E from April 7 to May 5, then the Eutelsat 9A/9B/10A at 9E and 10E from May 20 until Sept 23, at times coming well within 10 km", *Twitter*, October 4, 2023, <https://twitter.com/brianweeden/status/1709622408288543076>.

42. Michael Clonts, "Espionage in Orbit: Satellite or Spy?," *Kratos*, April 17, 2023, <https://www.kratosdefense.com/constellations/articles/espionage-in-orbit-satellite-or-spy>.
43. Bart Hendrickx, "Olimp and Yenisi-2: Russia's secretive eavesdropping satellites (part 1)," *The Space Review*, November 20, 2023, <https://www.thespacereview.com/article/4696/1>.
44. Stephen Clark, "It's hunting season in orbit as Russia's killer satellites mystify skywatchers," *Ars Technica*, July 11, 2025, <https://arstechnica.com/space/2025/07/its-hunting-season-in-orbit-as-russias-killer-satellites-mystify-sky-watchers/>.
45. s2a Systems, "A few images of COSMOS 2589 and the UNID that is moving away from it, which we noticed for the first time in the night from June 21 to 22," *Bluesky*, June 26, 2025, <https://web-cdn.bsky.app/profile/s2a-systems.bsky.social/post/3lsk37tgr22d>.
46. Greg Gillinger, "Updates!," *The Integrity Flash* Issue 126, August 3, 2025, <https://isruniversity.com/2025/08/04/issue-126/>; s2a Systems, "A few images of COSMOS 2589 and the UNID that is moving away from it, which we noticed for the first time in the night from June 21 to 22," *ibid*.
47. Gillinger, "Updates!" *ibid*.
48. "Video: Cosmos 2589 and Cosmos 2590," *COMSPOC*, September 15, 2025, <https://www.comspoc.com/videos/cosmos-2589-cosmos-2590#content-detail>.
49. Greg Gillinger, "Russia update: Cosmos 2589 and Object C," *The Integrity Flash* Issue 127, August 18, 2025, <https://isruniversity.com/2025/08/18/issue-127/#section-6>; Greg Gillinger, "Russia: Object C Renews RPO with Cosmos 2589," *The Integrity Flash* Issue 128, August 30, 2025, <https://isruniversity.com/2025/09/02/issue-128/>.
50. Greg Gillinger, "Cosmos 2589 Moving to GEO," *The Integrity Flash* Issue 135, December 8, 2025, <https://isruniversity.com/2025/12/08/the-integrity-flash-issue-135/#section1>.
51. Greg Gillinger, "Update: Cosmos 2589 Inching Toward GEO," *The Integrity Flash* Issue 137, January 19, 2026, <https://isruniversity.com/2026/01/19/issue-137/#tb12>.



525 Zang Street, STE. D
Broomfield, CO 80021 USA
v: + 1 303 554 1560

1779 Massachusetts Ave. NW
Washington, DC 20036 USA
v: + 1 202 568 6212

e: info@swfound.org